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Advanced Computer Architecture

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Prof. Marco Santambrogio

Name
Last Name

The student has to choose 2 out of the 3 exercises.

If all the 3 exercises will be completed, just the 2 with the lowest score will be considered.

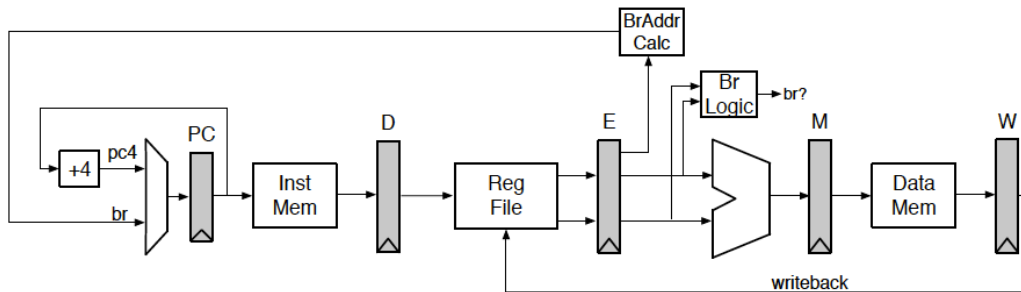
This is an open book exam, which means the students can use all their notes (we do not care of the format) but they cannot share their exams with anyone during the test.

At the end of the exam, when the professor will call it, each student has to upload 1 PDF file (named using the “codice persona” – CodicePersona.pdf) using the Microsoft Form share via the Zoom chat at the beginning of the exam.

In the first page of the uploaded PDF, the professor has to find the name and the surname of the student.

Problem 1 (50%)	
Problem 2 (50%)	
Problem 3 (50%)	
Total (100%)	

Problem 1



Consider the above 5 stages pipelined architecture.

Question 1: Indicate what kind of hazards can occur in this pipeline, motivating your answer

Consider the following executed code.

I1: ADDI R4, R0, R1

I2: ADDI R5, R0, R2

I3: BEQ R4, R5, 0x1000

I4: LW R7, 4(R6)

I5: OR R5, R7, R5

I6: SW R5, 8(R6)

Question 2: Complete the table and define how many clock cycles are needed.

	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	T11	T12	T13	T14	T15	T16	T17	T18	T19	T20
I1																				
I2																				
I3																				
I4																				
I5																				
6																				

Question 3:

Provide the architectural solutions to improve the performance for this code for the data hazards, motivating your answers and compute the performance improvement you can obtain.

Considering the control hazard, what improvement can you provide with static branch prediction?

How can you provide precise exceptions in this pipeline?

Problem 2

Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, write-allocate, and invalidation of other caches on write (instead of updating the value in the other caches).

Question: Is the proposed solution right? If no, please provide the right one.

NOTE: you can recreate the table in excel, you can draw it on paper, you can create a "ASCII" version of it... the only thing is that, at the end, the information has to be clear and not ambiguous.

Time	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 1	Exclusive (1)	Invalid (0)	Yes	Yes
1	P2: read block 0	Exclusive (1)	Exclusive (0)	Yes	Yes
2	P1: read block 0	Shared (0)	Shared (0)	Yes	Yes
3	P2: write block 0	Invalid (0)	Modified (0)	No	Yes
4	P1: read block 1	Exclusive (1)	Modified (0)	No	Yes
5	P2: read block 1	Shared (1)	Shared (1)	Yes	Yes
6	P1: read block 1	Shared (1)	Shared (1)	Yes	Yes
7	P2: write block 1	Invalid (1)	Modified (1)	Yes	No
8	P1: read block 0	Exclusive (0)	Modified (1)	Yes	No
9	P2: read block 1	Exclusive (0)	Modified (1)	Yes	No
10	P2: write block 1	Exclusive (0)	Modified (1)	Yes	No
11	P1: read block 1	Shared (1)	Shared (1)	Yes	Yes
12	P2: read block 1	Shared (1)	Shared (1)	Yes	Yes
13	P1: write block 1	Invalid (1)	Modified (1)	Yes	No

Problem 3

Assume that the following code has been executed on a CPU with SCOREBOARD.

	Issue	Read Op	Exec Co.	Write R.
I1: LD F6 32+ R2	1	2	7	8
I2: ADDD F2 F6 F4	2	9	11	12
I3: MULTD F0 F6 F2	3	13	43	44
I4: SUBD F6 F4 F6	4	9	11	14
I5: ADDD F0 F12 F2	45	46	48	49

A. List all the possible conflicts in the code.

B. Is there a "configuration" that can respect the shown execution?
How many units? Which kind? What latency?

C. If the answer to (B) was NO, please, write the right one supposing to have:

- 1 Memory unit, with a latency of 5cc
- 2 FP add/sub units, with a latency of 2cc
- 1 FM mult/div unit, with a latency of 30cc